

AP CALCULUS AB – UNIT 4

4.2 Straight Line Motion



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Relate position, velocity, speed, and acceleration using derivatives.
- Determine direction of motion, rest times, and changes in speed.
- Analyze motion from formulas, graphs, and tables.
- Calculate displacement and total distance over intervals using turning times.

Keys: Essential Knowledge and Vocabulary

Position	$s(t)$ locates a particle on a line; units are distance units.
Velocity	$v(t) = s'(t)$; its sign determines direction.
Speed	$ v(t) $; speed is never negative.
Acceleration	$a(t) = v'(t) = s''(t)$; it measures change in velocity.
Speeding up	Velocity and acceleration have the same sign.
Slowing down	Velocity and acceleration have opposite signs.
Displacement	$s(b) - s(a)$; total distance adds absolute position changes across direction reversals.

Explanation, Strategy, and Common Errors

- Find velocity and acceleration before solving sign questions.
- A particle moves right when $v > 0$, left when $v < 0$, and is at rest when $v = 0$.
- Use a sign chart for both v and a ; do not infer speeding up from $a > 0$ alone.
- To compute total distance without integration, locate all times where velocity changes sign and add absolute changes in position.
- A zero velocity can be a stop without a direction reversal if the sign of velocity does not change.

Solved Example: Direction and rest

For $s(t) = t^3 - 6t^2 + 9t$, find the velocity and the times the particle is at rest.

Answer and Reasoning

$v(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$. The particle is at rest at $t = 1$ and $t = 3$.

Solved Example: Speeding up

For $v(t) = t^2 - 4t + 3$ and $a(t) = 2t - 4$, determine where the particle speeds up on $0 < t < 5$.

Answer and Reasoning

Velocity is positive on $(0, 1) \cup (3, 5)$ and negative on $(1, 3)$; acceleration is negative on $(0, 2)$ and positive on $(2, 5)$. Same signs occur on $(1, 2) \cup (3, 5)$.

Solved Example: Total distance

A particle has $s(t) = t^3 - 3t^2$ for $0 \leq t \leq 4$. Find total distance.

Answer and Reasoning

$v = 3t(t - 2)$ changes sign at $t = 2$. Positions: $s(0) = 0$, $s(2) = -4$, $s(4) = 16$. Distance = $|-4 - 0| + |16 - (-4)| = 24$.

Solved Example: Motion from a table

Using the shared table, describe motion at $t = 1$ and $t = 4$.

Answer and Reasoning

At $t = 1$, $v = 2 > 0$ and $a = -2 < 0$: moving right and slowing down. At $t = 4$, $v = -4$ and $a = -2$: moving left and speeding up.

Shared Representation / Data

t	0	1	2	3	4	5	6
$s(t)$	4	7	8	7	4	-1	-8
$v(t)$	4	2	0	-2	-4	-6	-8
$a(t)$	-2	-2	-2	-2	-2	-2	-2

AP Exam Language

For every applied derivative, include the sign, units, input value, and a complete interpretation. In related-rates work, differentiate before substituting. For L'Hôpital problems, verify an allowed indeterminate form first.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. If $s(t) = 5t^2 - 3t + 1$, find $v(t)$ and $a(t)$.

2. If $v(4) = -7$, state the direction and speed at $t = 4$.

3. A particle has $v(t) = t - 6$. When is it at rest?

4. If $v(2) > 0$ and $a(2) > 0$, is the particle speeding up or slowing down?

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. For $s(t) = t^3 - 9t$, find all rest times for $t \geq 0$.

6. For $v(t) = (t - 2)(t - 5)$, determine intervals of rightward motion.

7. For $v(t) = t^2 - 6t + 5$, determine where velocity is increasing.

8. A particle moves from $s(0) = 3$ to $s(5) = -2$. Find displacement.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. For $s(t) = t^3 - 6t^2 + 9t$, determine intervals of right and left motion for $t > 0$.

10. For the same position on $0 \leq t \leq 4$, find displacement and total distance.

11. If $v(c) = 0$ and $a(c) > 0$, explain why a direction reversal is not automatic from these two values alone.

12. A velocity graph is below the axis but increasing. Describe motion.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. For $v(t) = t^3 - 3t^2 - 9t + 27$, determine all intervals on which speed increases.

14. A particle has $s(t) = \frac{1}{4}t^4 - 2t^2$ on $[-3, 3]$. Find all local extrema of position.

15. A particle has $v(t) = 0$ for an interval $[2, 3]$. What does this imply about s there?

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Construct a differentiable velocity function that is zero at $t = 1$ but does not reverse direction there.

17. Suppose $s'(t) = |t - 2|$. Is s twice differentiable at $t = 2$? Explain.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. A particle is slowing down when

- (A) $v > 0, a > 0$ (B) $v < 0, a < 0$
(C) v and a have opposite signs (D) $a = 0$

2. If $s'(4) = -3$, then at $t = 4$ the particle

- (A) moves right at speed 3 (B) moves left at speed 3
(C) has acceleration -3 (D) is at rest

3. For $s(t) = t^2 - 8t$, the particle is at rest at

- (A) $t = 0$ (B) $t = 2$
(C) $t = 4$ (D) $t = 8$

4. A velocity graph is negative with negative slope. The particle is

- (A) moving left and speeding up (B) moving left and slowing down
(C) moving right and speeding up (D) at rest

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For $s(t) = t^3 - 4t$, the velocity at $t = 1.7$ is closest to

- (A) 1.33 (B) 4.67
(C) 6.27 (D) 8.67

6. For $s(t) = t^3 - 3t^2$ on $0 \leq t \leq 4$, total distance is

- (A) 16 (B) 20
(C) 24 (D) 28

AP-Style Free-Response Practice

No Calculator

A particle moves along the x -axis with position $s(t) = t^3 - 6t^2 + 9t + 2$ for $0 \leq t \leq 5$.

(a) Find the velocity and acceleration. [2 points]

(b) Determine intervals on which the particle is speeding up. [3 points]

(c) Find total distance traveled on $[0, 5]$. [3 points]

Calculator Permitted

Velocity data for a particle are collected near $t = 4$ seconds: $v(3.9) = -1.42$, $v(4.0) = -1.20$, $v(4.1) = -0.94$.

(a) Estimate $a(4)$ using a symmetric difference. [2 points]

(b) At $t = 4$, is the particle speeding up or slowing down? Justify. [2 points]

(c) Estimate $v(4.25)$ using local linearization. [2 points]